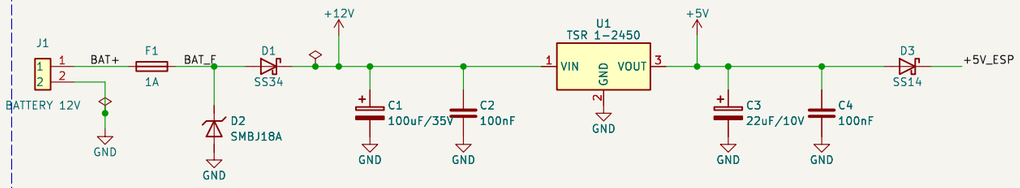
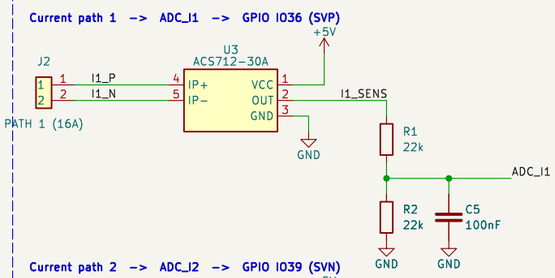
POWER SUPPLY 12 V –> 5 V

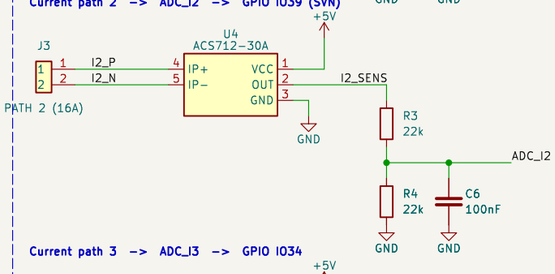


CURRENT SENSORS ACS712-30A

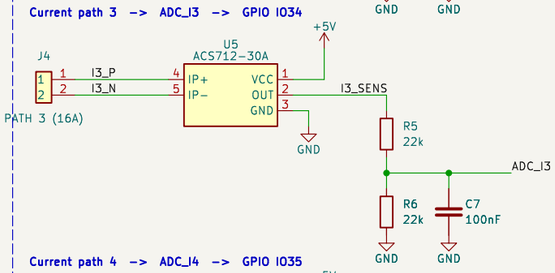
Current path 1 –> ADC_I1 –> GPIO IO36 (SVP)



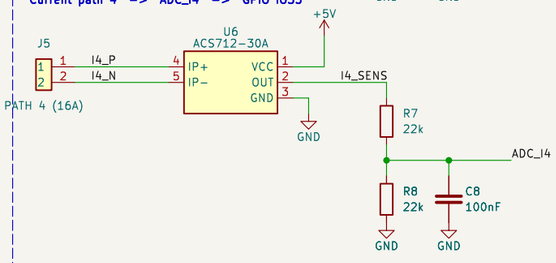
Current path 2 –> ADC_I2 –> GPIO IO39 (SVN)



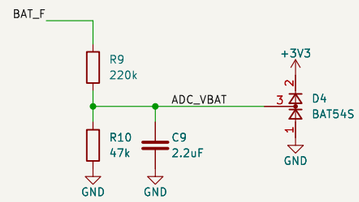
Current path 3 –> ADC_I3 –> GPIO IO34



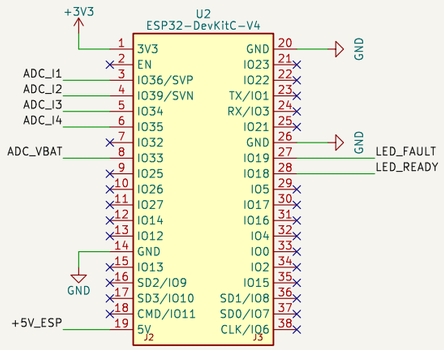
Current path 4 –> ADC_I4 –> GPIO IO35



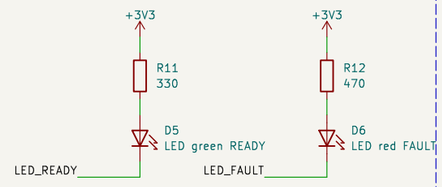
BATTERY VOLTAGE (2.11 V @ 12 V)



CONTROLLER ESP32-DevKitC V4



STATUS LEDs (active LOW)



NOTES

1. Supply: 12 V battery –> F1 (1 A) –> D1 SS34 (reverse polarity) –> U1 TRACO TSR 1-2450 (5 V / 1 A).
   D2 SMBJ18A clamps transients from the wind controller; on reverse polarity D2 conducts and F1 opens.
2. D3 SS14 prevents back-feeding when the DevKitC is powered via USB at the same time (ESP 5V pin approx. 4.7 V).
3. U3..U6 ACS712-30A breakout modules (on-board decoupling assumed). Output ratiometric to +5V.
   Divider 22k/22k –> ADC pin approx. 1.25 V at 0 A (prototype measured 1.34 V, firmware does zero calibration).
4. High-current paths J2..J5 (IP+/IP-): up to 16 A, 2.5 mm2 wiring, own fuse per path (not shown).
5. Battery divider R9/R10 220k/47k taps BAT_F (before D1) = true battery voltage. D4 BAT54S clamps ADC to -0.3..3.6 V.
6. LEDs active LOW: GPIO sinks current. GPIO18 = READY (green), GPIO19 = FAULT (red).
7. All analog inputs on ADC1 (not affected by WiFi). U2 pins 1-19 = header J2 (left), 20-38 = header J3 (right).
8. This ESP32 has no ADC eFuse calibration (ADC_VREF = 0) –> linear correction in firmware wind_monitor.ino.

Firmware: wind_monitor.ino
ESP32-DevKitC V4, 4x ACS712-30A, battery voltage, status LEDs
Tom

Sheet: /
File: wind_monitor.kicad_sch

Title: Wind Monitor – ESP32 Signal Test

Size: A3	Date: 2026-09-30	**Rev: 0.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1